

COMPLEX ENGINEERING

PROBLEM

STATEMENT

A trunnion has to be cooled before it is shrink fitted into a steel hub.

The equation gives temp. T_f to which the trunnion has to be cooled to obtain the desired contraction is given by

$$F(T_f) = -0.50598 \times 10^{-10} T_f^3 + 0.38292 \times 10^{-7} T_f^2 + 0.74363 \times 10^{-4} T_f + 0.88318 \times 10^2 = 0$$

Use Newton Raphson Method of finding roots of equations to find the temp. T_f to which the trunnion has to be cooled conduct three iterations to estimate the roots of above equation.

SOLUTION

$$F(T_f) = -0.50598 \times 10^{-10} T_f^3 + 0.38292 \times 10^{-7} T_f^2 + 0.74363 \times 10^{-4} T_f + 0.88318 \times 10^{-2}$$

$$F'(T_f) = 1.51794 \times 10^{-10} T_f^2 + 0.76584 \times 10^{-7} T_f + 0.74363 \times 10^{-4}$$

$$\text{Let, } T_{f_0} = -100$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f(-100) = -0.50598 \times 10^{-10} (-100)^3 + 0.38292 \times 10^{-7} (-100)^2 + 0.74363 \times 10^{-4} (-100) + 0.883 \times 10^{-2}$$

$$f(-100) = 1.8290 \times 10^{-3}$$

$$f'(-100) = 1.5179 \times 10^{-10} (-100)^2 + 0.76582 \times 10^{-7} (-100) + 0.74363 \times 10^{-4}$$

$$f'(-100) = 6.5187 \times 10^{-5}$$

Now,

$$T_{f_1} = -100 - \frac{(1.8290 \times 10^{-3})}{6.5187 \times 10^{-5}}$$

$$T_{f_1} = -128.06$$

$$F(-128.06) = -0.50598 \times 10^{-10} (-128.06)^3 + 0.38292 \times 10^{-7} (-128.06)^2 + 0.76584 \times 10^{-4} (-128.06) + 0.88318 \times 10^{-2}$$

 $10^{-7} T_f^2$

$$F(-128.06) = 4.3214 \times 10^{-5}$$

$$F'(-128.06) = -1.5179 \times 10^{-10} (-128.06)^2 + 0.76584 \times 10^{-7} (-128.06) + 0.74363 \times 10^{-4}$$

 T_f

$$F'(-128.06) = 6.2067 \times 10^{-5}$$

$$T_{f_1} = -128.06 - \frac{4.3214 \times 10^{-5}}{6.2067 \times 10^{-5}}$$

$$T_{f_1} = -128.75$$

 $0)^2$

$$F(-128.75) = -0.50598 \times 10^{-10} (-128.75)^3 + 0.38292 \times 10^{-7} (-128.75)^2 + 0.74363 \times 10^{-4} (-128.75) + 0.88318 \times 10^{-2}$$

 $100)$

$$F(-128.75) = 2.8002 \times 10^{-8}$$

$$F'(-128.75) = 1.5179 \times 10^{-10} (-128.75)^2 + 0.76584 \times 10^{-7} (-128.75) + 0.74363 \times 10^{-4}$$

$$F'(-128.75) = 6.1986 \times 10^{-5}$$

$$T_{f_3} = -128.75 - \frac{2.8002 \times 10^{-8}}{6.1986 \times 10^{-5}}$$

$$T_{f_3} = -128.75$$

Since, $T_{f_2} = T_{f_3} = -128.75$ (approx), the contraction is 128.75°C

CONCLUSION

The problem demonstrates how the Newton Raphson method is used for solving non-linear equations in an engineering context. The final result $T_f = 128.75$ ensures that the contraction of the steel hub matches the desired value, allowing fitting in the assembly process.

REAL WORLD USE

This problem finds practical use in mechanical engineering and manufacturing, especially in scenarios like interference fits like shrink fitting metal parts.

FIELD OF THE PROBLEM

The problem belongs to mechanical engineering

which deals with shrink fit techniques, material science which involve understanding thermal expansion and contraction and applied mathematics where numerical method like Newton Raphson are used to solve non-linear equations for real world application

Hence,

Calculus helps solve this problem by providing tools like derivatives and numerical approximation techniques. These are critical in analyzing and optimizing thermal expansion or contraction behaviour in engineering design ensuring precision and safety.

INTEGRATION

QUESTION

Evaluate the following.

$$\int_0^{\infty} \sqrt{x} e^{-x^3} dx$$

Solution

Let $U = (\sqrt{x^3})$

$$\frac{dU}{dx} = 3x^2 \Rightarrow dx = \frac{1}{3x^2} dU$$

$$\Rightarrow \sqrt{x} = U^{1/6}$$

$$\Rightarrow \int_0^{\infty} U^{1/6} \cdot e^{-U} \cdot \frac{1}{3x^2} dU$$

$$\Rightarrow \int_0^{\infty} U^{1/6} \cdot e^{-U} \cdot \frac{1}{3(U^{1/6})^2} dU$$

$$\Rightarrow \frac{1}{3} \int_0^{\infty} U^{1/6} \cdot U^{-2/3} \cdot e^{-U} dU$$

$$\Rightarrow \frac{1}{3} \int_0^{\infty} U^{-3/6} \cdot e^{-U} dU$$

$$\Rightarrow \frac{1}{3} \int_0^{\infty} U^{-1/2} \cdot e^{-U} dU$$

$$\Rightarrow \frac{1}{3} \int_0^{\infty} U^{1/2-1} \cdot e^{-U} dU$$

$$\Rightarrow \frac{1}{3} \left[\frac{1}{2} \right]$$

$$\Rightarrow \frac{\pi}{3}$$

Answer

2

$$\int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx$$

Solution

$$\text{Let } U = \sqrt[3]{x} ; U^{3/2} = \sqrt{x}$$

$$\Rightarrow U^3 = x \Rightarrow \frac{dx}{dU} = 3U^2$$

$$\Rightarrow \frac{dU}{dx} = \frac{1}{3} \cdot \frac{1}{(U)^2}$$

$$\Rightarrow dx = 3U^2 dU$$

$$3 \int_0^{\infty} e^{-U} U^{3/2} \cdot U^2 dU$$

$$\Rightarrow 3 \int_0^{\infty} e^{-U} U^{7/2} dU$$

$$= 3 \int_0^{\infty} e^{-u} \cdot u^{7/2} du$$

$$= 3 \int_0^{\infty} e^{-u} u^{\frac{9}{2}-1} du$$

$$= 3 \left[\frac{9}{2} \right]$$

$$= 3 \left[\frac{7+1}{2} \right]$$

$$= \frac{3 \times 7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$$

$$= \frac{315 \sqrt{\pi}}{16}$$

Answer

3

$$\int_0^1 x^3 (1-x)^{4/3} dx$$

Solution

$$\text{Let } U = x^3$$

$$= \int_0^1 (U)^3 (1-U)^{4/3} dx$$

$$= \int_0^1 (U)^{4-1} (1-U)^{\frac{4}{3}-1} dU$$

$$= \frac{1}{6} B(4, 7/3)$$

$$= \frac{\sqrt{4} \sqrt{7/3}}{\sqrt{19/3}}$$

$$= \frac{3! \times 4/3!}{16/3 \times 13/3 \times 10/3 \times 7/3 \times 4/3!}$$

$$= \frac{6 \times 81}{14560}$$

$$= \frac{243}{7280}$$

Answer

$$\int_0^{\infty} \sqrt{x} e^{-x} dx$$

Solution

$$\text{Let } U = x ; \quad \sqrt{U} = \sqrt{x}$$

$$= \int_0^{\infty} e^{-u} \sqrt{u} du$$

$$= \int_0^{\infty} e^{-u} u^{\frac{3}{2}-1} du$$

$$= \left[\frac{3}{2} \right]$$

$$= \frac{3}{2} \left[\frac{1}{2} \right] \Rightarrow \frac{3}{2} \times \frac{1}{2} \sqrt{\pi}$$

$$= \frac{3}{4} \sqrt{\pi}$$

Answer

DOUBLE INTEGRATION

QUESTION

evaluate the following;

$$\int_1^3 \int_0^2 (y^3 - xy) \, dy \, dx$$

Solution

$$= \int_1^3 dx \int_0^2 (y^3 - xy) \, dy$$

$$= \int_1^3 dx \left[\frac{y^4}{4} - \frac{xy^2}{2} \right] \Big|_0^2$$

$$= \int_1^3 dx \left[\frac{(2)^4}{4} - \frac{x(2)^2}{2} - 0 \right]$$

$$= \int_1^3 dx [4 - 2x]$$

$$= \int_1^3 [4 - 2x] \, dx$$

$$= \left[\frac{4(3) - 2(3)^2}{2} \right] - \left[\frac{4(1) - 2(1)^2}{2} \right]$$

$$= (12 - 9) - (4 - 1)$$

$$= 0$$

Answer

$$\int_1^2 \int_0^\pi (3 + \sin \theta) d\theta dr$$

Solution

$$\begin{aligned} & \int_1^2 dr \int_0^\pi (3 + \sin \theta) d\theta \\ &= \int_1^2 dr [3\pi + 1 + 1] \\ &= \int_1^2 dr [3\pi + (-\cos \theta)] \Big|_0^\pi \\ &= \int_1^2 dr [3\pi + 1 + 1] \\ &= \int_1^2 (3\pi + 2) dr \\ &= \left[3\pi r + 2r \right] \Big|_1^2 \\ &= [6\pi + 4 - 3\pi - 2] \\ &= 3\pi + 2 \end{aligned}$$

Answer

INTEGRAL CALCULUS

Evaluate the following.

Q 1 B

$$\int_0^{\pi/2} \cos^7 x \, dx$$

Solution

$n = \text{odd} = 7$

$$\int_0^{\pi/2} \cos^n x \, dx = \frac{n-1}{n} \times \frac{n-3}{n-2} \times \frac{n-5}{n-4} \dots \times \frac{2}{3}$$

apply formula

$$\int_0^{\pi/2} \cos^7 x \, dx = \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3}$$

$$\int_0^{\pi/2} \cos^7 x \, dx = \frac{16}{35}$$

Answer

Q 2 B

Solution

$$\int \sec^6 x \, dx$$

$\Rightarrow n = \text{even} = 6$

$$\therefore \int \sec^n x \, dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2} x \, dx$$

$$\int \sec^6 x \, dx = \frac{\tan x \sec^4 x}{5} + \frac{4}{5} \int \sec^4 x \, dx$$

again apply formula

$$\int \sec^6 x \, dx = \frac{1}{5} \tan x \sec^4 x + \frac{4}{5} \left[\frac{\tan x \sec^2 x}{3} + \frac{2}{3} \int \sec^2 x \, dx \right]$$

$$\int \sec^6 x \, dx = \frac{1}{5} \tan x \sec^4 x + \frac{4}{15} \tan x \sec^2 x + \frac{8}{15} \tan x + C$$

Answer

3

$$\int \tan^6 x \, dx$$

Solution

The reduction formula is given by

$$\therefore \int \tan^n x \, dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x \, dx$$

apply formula

$$\Rightarrow \int \tan^6 x \, dx = \frac{\tan^5 x}{5} - \int \tan^4 x \, dx$$

again apply formula

$$\Rightarrow \int \tan^6 x dx = \frac{1}{5} \tan^5 x - \left\{ \frac{\tan^3 x}{3} - \int \tan^2 x dx \right\}$$

again apply formula

$$\Rightarrow \int \tan^6 x dx = \frac{1}{5} \tan^5 x - \frac{1}{3} \tan^3 x + \left\{ \tan x - \int \tan^0 x dx \right\}$$

$$\Rightarrow \int \tan^6 x dx = \frac{1}{5} \tan^5 x - \frac{1}{3} \tan^3 x + \tan x - x + C$$

Answer

Q 4 B

$$\int \sin^6 x \cos^8 x dx$$

Solution

→ putting $p=6$ and $q=8$ in the formula;

$$= \int \sin^6 x \cos^8 x dx = \frac{1}{6+8} \cos^{8+1} x \sin^6 x + \frac{6-1}{6-8} \int \sin^{6-2} x \cos^8 x dx$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cdot \cos^9 x \sin^6 x + \frac{5}{14} \int \sin^4 x \cos^8 x dx$$

→ putting $p=4$ and $q=8$

$$= \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^6 x + \frac{5}{14} \left[\frac{-1}{4+8} \cdot (\cos^{8+1} x \sin^4 x) + \frac{4-1}{4+8} \int \sin^{4-2} x \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x + \frac{5}{14} \left[\frac{-1}{12} \cos^9 x \sin^3 x + \frac{3}{12} \int \sin^4 x \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x + \frac{5}{14} \left[\frac{-1}{12} \cos^9 x \sin^3 x + \frac{3}{12} \int \sin^4 x \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x + \frac{5}{56} \int \sin^4 x \cos^8 x dx$$

→ put $p=2$ and $q=8$ now;

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x + \frac{5}{56} \left[\frac{-1}{2+8} (\cos^{8+1} x \sin^{2-1} x) + \frac{2-1}{2+8} \int \sin^{2-2} x \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x + \frac{5}{56} \left[\frac{-1}{10} \cos^9 x \sin x + \frac{1}{10} \int \sin^0 x \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x + \frac{5}{56} \left[\frac{-1}{10} \cos^9 x \sin x + \frac{1}{10} \int \cos^8 x dx \right]$$

$$\Rightarrow \int \sin^6 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^7 x \sin^3 x - \frac{1}{112} \cos^5 x \sin x + \frac{1}{112} \int \cos^8 x dx$$

$$\therefore \int \cos^n x dx = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} \int \cos^{n-2} x dx$$

→ putting $n=8$

$$\Rightarrow \int \sin^4 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^7 x \sin^3 x - \frac{1}{112} \cos^5 x \sin x + \frac{1}{112} \left[\frac{1}{8} (\sin x \cos^8 x) + \frac{8-1}{8} \int \cos^6 x dx \right]$$

→ putting $n=6$

$$\Rightarrow \int \sin^2 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^7 x \sin^3 x - \frac{1}{112} \cos^5 x \sin x + \frac{1}{896} \sin x \cos^7 x + \frac{13}{128} \left[\frac{1}{6} (\sin x \cos^6 x) + \frac{6-1}{6} \int \cos^4 x dx \right]$$

→ put $n=4$

$$\Rightarrow \int \sin^2 x \cos^8 x dx = \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^7 x \sin^3 x - \frac{1}{112} \cos^5 x \sin x + \frac{1}{896} \sin x \cos^7 x + \frac{1}{168} \sin x \cos^5 x + \frac{5}{768} \left[\frac{1}{4} (\sin x \cos^4 x) + \frac{4-1}{4} \int \cos^2 x dx \right]$$

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→ put $n=2$ now;

$$\begin{aligned} \Rightarrow \int \sin^6 x \cos^8 x dx &= \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x - \frac{1}{112} \cos^9 x \\ &\quad \sin x + \frac{1}{896} \sin x \cos^7 x + \frac{1}{768} \sin x \cos^5 x + \\ &\quad \frac{5}{3072} \sin x \cos^3 x + \frac{5}{1024} \left[\frac{1}{2} (\sin x \cos^{2-1} x) + \right. \\ &\quad \left. \frac{2-1}{2} \int \cos^{2-2} x dx \right] \end{aligned}$$

$$\begin{aligned} \Rightarrow \int \sin^6 x \cos^8 x dx &= \frac{-1}{14} \cos^9 x \sin^5 x - \frac{5}{168} \cos^9 x \sin^3 x \\ &\quad - \frac{1}{112} \cos^9 x \sin x + \frac{1}{896} \sin x \cos^7 x + \\ &\quad \frac{1}{768} \sin x \cos^5 x + \frac{5}{3072} \sin x \cos^3 x \\ &\quad + \frac{5}{2048} \sin x \cos x + \frac{5}{2048} x + C \end{aligned}$$

Answer

N-R METHOD

QUESTION

Use Newton Raphson Method to compute

$$\sqrt[3]{1}$$

Sq. root of 27

Solution

Let x be the sq. rt of 27

$$x = \sqrt{27}$$

$$x^2 = 27$$

$$x^2 - 27 = 0$$

$$f(x) = x^2 - 27$$

$$f'(x) = 2x$$

$$x_0 = 5$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 5 - \frac{(-2)}{10}$$

$$x_1 = 5.25$$

$$f(5.25) = 0.5625, \quad f'(5.25) = 10.5$$

$$x_2 = 5.25 - \frac{0.5625}{10.5}$$

$$x_2 = 5.19642$$

$$f(5.196) = -1.584 \times 10^{-3} ; f'(5.196) = 10.392$$

$$x_3 = 5.196 + \frac{1.584 \times 10^{-3}}{10.392}$$

$$x_3 = 5.19615$$

$$x_2 = x_3 = 5.196 \text{ (approx)}$$

Answer

2

Sq. rt of 8

Solution

Let x be the sq. rt of 8

$$x = \sqrt{8}$$

Sq. rt on b.s

$$x^2 = 8$$

$$x^2 - 8 = 0$$

$$f(0) = -8$$

$$f(1) = -7$$

$$f(3) = 1$$

$$f(2.5) = -1.1$$

$$f'(3) = 6$$

$$f(x) = x^2 - 8$$

$$f'(x) = 2x$$

$$x_0 = 3$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 3 - \frac{1}{6}$$

$$x_1 = 2.83333$$



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$$f(x_1) = 0.02775 ; f'(x_1) = 5.6666$$

$$x_2 = 2.83333 - \frac{0.02775}{5.6666}$$

$$x_2 = 2.82840$$

$$f(x_2) = -1.5344 \times 10^{-4} ; f'(x_2) = 5.6568$$

$$x_3 = 2.82840 + \frac{1.5344 \times 10^{-4}}{5.6568}$$

$$x_3 = 2.82842$$

Hence,

$$x_2 = x_3 = 2.8284 \text{ (approx)}$$

Answer

Q 3 B

Cube root of 18

Solution

Let x be the cube root of 18

$$x = \sqrt[3]{18}$$

$$x = (18)^{1/3}$$

$$f(0) = -18$$

taking cube on b.s

$$f(2) = -10$$

$$x^3 = 18$$

$$f(3) = 9$$

$$x^3 - 18 = 0$$

$$f'(3) = 27$$

$$f(x) = 3x^2$$

$$\therefore x_0 = 3$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 3 - \frac{9}{27}$$

$$x_1 = 2.66666$$

$$f(x_1) = 0.96282 \quad ; \quad f'(x_1) = 21.33322$$

$$x_2 = 2.66666 - \frac{0.96282}{21.33322}$$

$$x_2 = 2.62152$$

$$f(x_2) = 0.01604 \quad ; \quad f'(x_2) = 20.6171$$

$$x_3 = 2.62152 - \frac{0.01604}{20.6171}$$

$$x_3 = 2.62074$$

$$f(x_3) = -2.87274 \times 10^{-5} \quad ; \quad f'(x_3) = 20.60483$$

$$x_4 = 2.62074 + \frac{2.87274 \times 10^{-5}}{20.60483}$$

$$x_4 = 2.62074$$

Hence,

$$x_3 = x_4 = 2.62074$$

Answer

Q 4 B

5th root of 17

Solution

Let x be the 5th root of 17

$$x = \sqrt[5]{17}$$

$$x = (17)^{1/5}$$

$$x^5 = 17$$

$$x^5 - 17 = 0$$

$$\therefore x_0 = 1.7$$

$$F(x) = x^5 - 17$$

$$F'(x) = 5x^4$$

$$\therefore x_1 = x_0 - \frac{F(x)}{F'(x)}$$

$$x_1 = 1.7 + \frac{2.8043}{41.7605}$$

$$x_1 = 1.76715$$

$$F(x_1) = 0.23324 \quad ; \quad F'(x_1) = 48.75999$$

$$x_2 = 1.76715 - \frac{0.23324}{48.75999}$$

$$x_2 = 1.76236$$

$$F(x_2) = 9.47871 \times 10^{-4} \quad ; \quad F'(x_2) = 48.23347$$

$$x_3 = 1.76236 - \frac{9.47871 \times 10^{-4}}{48.23347}$$

$$x_3 = 1.76234$$

Hence,

$$x_2 = x_3 = 1.7623 \quad (\text{approx})$$

The 5th root of 17 is 1.7623

Answer

Q 5 B

$$\sqrt[3]{13}$$

Solution

$$\text{Let } x = \sqrt[3]{13}$$

$$x^3 = 13$$

$$x^3 - 13 = 0$$

$$f(x) = x^3 - 13$$

$$f'(x) = 3x^2$$

$$\therefore x_0 = 2.4$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 2.4 - \frac{0.824}{11.52}$$

$$x_1 = 2.32847$$

$$f(x_1) = -0.37556 \quad ; \quad f'(x) = 16.26531$$

$$x_2 = 2.32847 + \frac{0.37556}{16.26531}$$

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$$x_2 = 2.35155$$

$$f(x_2) = 3.57156 \times 10^{-3} ; f'(x_2) = 16.58936$$

$$x_3 = 2.35155 - \frac{3.5716 \times 10^{-3}}{16.58936}$$

$$x_3 = 2.35133$$

Hence,

$$x_2 = x_3 = 2.351 \text{ (approx)}$$

The $\sqrt[3]{13}$ is equals to 2.351

Answer

Q 6 B

$$\sqrt{125}$$

Solution

$$\text{Let } x = \sqrt{125}$$

$$x^2 = 125$$

$$x^2 - 125 = 0$$

$$f(0) = -125$$

$$f(11) = -4$$

$$f(11) = 22$$

$$f(x) = x^2 - 125$$

$$f'(x) = 2x$$

$$\therefore x_0 = 11$$

$$\therefore x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 11 + \frac{4}{22}$$

$$x_1 = 11.18181$$

$$F(x_1) = 0.32874 \quad ; \quad F'(x_1) = 22.36362$$

$$x_2 = 11.18181 - \frac{0.32874}{22.36362}$$

$$x_2 = 11.16711$$

$$F(x_2) = -0.29565 \quad ; \quad F'(x_2) = 22.33422$$

$$x_3 = 11.16711 + \frac{0.29565}{22.3422}$$

$$x_3 = 11.8033$$

$$F(x_3) = -2.210911 \times 10^{-4} \quad ; \quad F'(x_3) = 22.36066$$

$$x_4 = 11.8033 + \frac{2.210911 \times 10^{-4}}{22.36066}$$

$$x_4 = 11.8033$$

Hence,

$$x_3 = x_4 = 11.8033$$

Therefore, sq. root of 125 is equal to 11.8033

Answer

COMPLEX NUMBERS

QUESTION

Find the modulus and argument of the complex number

$$\sqrt{1}$$

Solution

$$\frac{1 + 2i}{1 - (1 - i)^2}$$

$$= \frac{1 + 2i}{1 - (1 + i^2 - 2i)}$$

$$= \frac{1 + 2i}{1 - 1 - i^2 + 2i}$$

$$= \frac{1 + 2i}{-i^2 + 2i}$$

$$= \frac{1 + 2i}{-(-1) + 2i}$$

$$= \frac{1 + 2i}{1 + 2i}$$

$$= 1$$

Let $z = 1$

$\Rightarrow z = 1 + 0i$

MODULUS

$$|z| = \sqrt{a^2 + b^2} = \sqrt{(1)^2 + (0)^2} = 1 \text{ unit}$$

ARGUMENT

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{0}{1}\right) = \tan^{-1}(0) = 0$$

Hence, modulus is 1 unit and
argument is 0.

QUESTION

If $z = x + yi$, prove that $\frac{z + \bar{z}}{\bar{z} - z} = 2 \left(\frac{x^2 - y^2}{x^2 + y^2} \right)$

Solution

$$z = x + yi, \quad \bar{z} = x - yi$$

Taking L.H.S

$$= \frac{x + yi}{x - yi} + \frac{x - yi}{x + yi}$$

$$= \frac{(x + yi)^2 + (x - yi)^2}{x^2 - (yi)^2}$$

$$= \frac{x^2 + 2xyi + y^2i^2 + x^2 - 2xyi + y^2i^2}{x^2 - y^2i^2}$$

$$= \frac{x^2 + y^2(-1) + x^2 + y^2(-1)}{x^2 - y^2(-1)}$$

$$= \frac{x^2 - y^2 + x^2 - y^2}{x^2 + y^2}$$

$$= \frac{2x^2 - 2y^2}{x^2 + y^2}$$

$$= \frac{2(x^2 - y^2)}{x^2 + y^2}$$

$$= R.H.S$$

Hence, proved

QUESTION

If $z = a\cos\theta + ia\sin\theta$, prove that $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$

$$= 2\cos 2\theta$$

PROOF

$$z = a\cos\theta + ia\sin\theta$$

$$\bar{z} = a\cos\theta - ia\sin\theta$$

Taking L.H.S

$$= \frac{a \cos \theta + i a \sin \theta}{a \cos \theta - i a \sin \theta} + \frac{a \cos \theta - i a \sin \theta}{a \cos \theta + i a \sin \theta}$$

$$= \frac{(a \cos \theta + i a \sin \theta)^2 + (a \cos \theta - i a \sin \theta)^2}{(a \cos \theta)^2 - (i a \sin \theta)^2}$$

$$= \frac{a^2 \cos^2 \theta + i^2 a^2 \sin^2 \theta - 2(a \cos \theta)(i a \sin \theta) + a^2 \cos^2 \theta + i^2 a^2 \sin^2 \theta - 2(a \cos \theta)(i a \sin \theta)}{a^2 \cos^2 \theta - i^2 a^2 \sin^2 \theta}$$

$$= \frac{2a^2 \cos^2 \theta + 2i^2 a^2 \sin^2 \theta}{a^2 \cos^2 \theta - i^2 a^2 \sin^2 \theta}$$

$$= \frac{2a^2 (\cos^2 \theta - \sin^2 \theta)}{a^2 (\cos^2 \theta + \sin^2 \theta)}$$

$$= 2 \cos 2\theta$$

$$= R.H.S$$

Proved

QUESTION

prove that $\left| \frac{z-1}{\bar{z}-1} \right| = 1$

TO PROVE THAT

Taking L.H.S

$$= \left| \frac{z-1}{\bar{z}-1} \right|$$

$$\Rightarrow z = x+yi$$

$$\bar{z} = x-yi$$

$$= \left| \frac{x+yi-1}{x-yi-1} \right|$$

$$= \frac{|x-1+iy|}{|x-1-iy|}$$

$$= \frac{\sqrt{(x-1)^2+(y)^2}}{\sqrt{(x-1)^2+(y)^2}}$$

$$= \frac{\sqrt{x^2-2x+1+y^2}}{\sqrt{x^2-2x+1+y^2}}$$

$$= \frac{1}{1}$$

$$= 1$$

$$= \text{R.H.S}$$

Proved

QUESTIONExpress $1 - \sqrt{2}i$ in polar coordinatesSolution

$$P(r, \theta)$$

$$z = 1 - \sqrt{2}i$$

$$|z| = \sqrt{a^2 + b^2} = \sqrt{(1 - \sqrt{2})^2 + (1)^2} = \sqrt{4 - 2\sqrt{2}} \text{ units}$$

$$\theta = \tan^{-1} \frac{b}{a} = \tan^{-1} \frac{-1}{(1 - \sqrt{2})} = -\frac{3\pi}{8}$$

Hence, the polar form is;

$$z = r \operatorname{cis} \theta$$

$$z = \sqrt{4 - 2\sqrt{2}} \operatorname{cis} \left(-\frac{3\pi}{8} \right)$$

Answer

QUESTION

Let $z_1 = 2-i$ and $z_2 = -2+i$, find

~~of~~ 1 ~~is~~

$$\operatorname{Re} \left[\frac{z_1 z_2}{\bar{z}_1} \right]$$

Solution

$$= \frac{(2-i)(-2+i)}{(2+i)}$$

$$= \frac{-4 + 2i + 2i + 1}{2+i}$$

$$= \frac{4i - 3}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{8i + 4 - 6 + 3i}{(2)^2 - (i)^2}$$

$$= \frac{11i - 2}{5}$$

$$= \operatorname{Re} \left[\frac{-2}{5} \right]$$

Answer

Q.

2

$$= \operatorname{Im} \left[\frac{1}{z_1 \bar{z}_2} \right]$$

Solution

$$= \frac{1}{(2-i)(2+i)}$$

$$= \frac{1}{-4 + 2i + 2i + i^2}$$

$$= \frac{1}{-4 - 1}$$

$$= -\frac{1}{5}$$

It has no imaginary part thus;

$$\operatorname{Im}[0]$$

Answer

DE MOIVRE'S THEOREM

QUESTION

Expand the following in series of sines or cosines of multiple of θ .

$$\cos^5 \theta \sin^7 \theta$$

$$\cos^5 \theta \sin^7 \theta$$

Solution

$$\therefore x = \cos n\theta + i \sin n\theta$$

$$\therefore x^n + \frac{1}{x^n} = 2 \cos n\theta \quad ; \quad \therefore x^n - \frac{1}{x^n} = 2i \sin n\theta$$

$\hookrightarrow \text{eq. 1} \qquad \qquad \qquad \hookrightarrow \text{eq. 2}$

$$(2 \cos \theta)^5 = \left(x + \frac{1}{x} \right)^5$$

$$= x^5 + 5x^4 \cdot \frac{1}{x} + \frac{5(4)}{2!} x^3 \cdot \frac{1}{x^2}$$

$$+ \frac{5(4)(3)}{3!} x^2 \cdot \frac{1}{x^3} + \frac{5(4)(3)(2)}{4!} x \cdot \frac{1}{x^4}$$

$$+ \frac{5(4)(3)(2)(1)}{5!} x^0 \cdot \frac{1}{x^5}$$

$$2^5 \cos^5 \theta = x^5 + 5x^3 + 10x + \frac{10}{x} + \frac{5}{x^3} + \frac{1}{x^5}$$

→ eq A

Now, for $\sin^7 \theta$;

$$(2i \sin \theta)^7 = \left(\frac{x - \frac{1}{x}}{x} \right)^7$$

$$= \frac{x^7 - 7x^5 + 21x^3 - 35x + 35}{x^7} - \frac{21}{x^5} + \frac{7}{x^3} - \frac{1}{x}$$

→ eq B

Now, Xing eq A and eq B

$$2^5 \cos^5 \theta \cdot 2^7 \sin^7 \theta i^7 = \left(x^5 + 5x^3 + 10x + \frac{10}{x} + \frac{5}{x^3} + \frac{1}{x^5} \right) \times$$

$$\left(\frac{x^7 - 7x^5 + 21x^3 - 35x + 35}{x^7} - \frac{21}{x^5} + \frac{7}{x^3} - \frac{1}{x} \right)$$

$$\rightarrow 2^5 \cdot 2^7 \cdot i^7 \cos^5 \theta \sin^7 \theta = x^{12} - 7x^{10} + 21x^8 - 35x^6 + 35x^4 - 21x^2$$

$$+ 7 - \frac{1}{x^2} + 5x^{10} - 35x^8 + 105x^6 - 175x^4$$

$$+ 17x^2 - 105 + \frac{35}{x^2} - \frac{5}{x^4} + 10x^8$$

$$- 70x^6 + 210x^4 - 350x^2 + 350$$

$$- \frac{210}{x^2} + \frac{70}{x^4} - \frac{10}{x^6} + 10x^6 - 70x^4$$

$$+ 210x^4 - 350 + \frac{350}{x^2} - \frac{210}{x^4}$$

$$+ \frac{70}{x^2} - \frac{10}{x^4} + 5x^4 - 35x^2$$

$$+ \frac{105}{x^2} - \frac{175}{x^4} + \frac{175}{x^4} - \frac{105}{x^6}$$

$$+ \frac{35}{x^8} - \frac{5}{x^{10}} + \frac{x^2 - 7 + 21}{x^2}$$

$$- \frac{35}{x^4} + \frac{35}{x^4} - \frac{21}{x^8} + \frac{7}{x^{10}} - \frac{1}{x^{12}}$$

$$= \frac{x^{12} - 1}{x^{12}} - \frac{2x^{10} - 2}{x^{10}} - \frac{4x^8}{x^8}$$

$$+ \frac{4}{x^8} + \frac{10x^6}{x^6} - \frac{10}{x^6} + \frac{5x^4}{x^4}$$

$$- \frac{20x^2 + 20}{x^2}$$

$$= 2i \sin 12\theta - 2(2i \sin 10\theta) - 4(2i \sin 8\theta) + 10(2i \sin 6\theta) + 5(2i \sin 4\theta) - 20(2i \sin 2\theta)$$

$$= 2i \{ \sin 12\theta - 2\sin 10\theta - 4\sin 8\theta + 10\sin 6\theta + 5\sin 4\theta - 20\sin 2\theta \}$$

$$\cos^5 \theta \cdot \sin^7 \theta = \frac{2i}{2^5 \cdot 2^7 \cdot i^7} \{ \sin 12\theta - 2\sin 10\theta$$

$$- 4\sin 8\theta + 10\sin 6\theta + 5\sin 4\theta - 20\sin 2\theta \}$$

$$= \frac{1}{2048 \cdot i \cdot i} \{ \sin 12\theta - 2\sin 10\theta - 4\sin 8\theta + 10\sin 6\theta + 5\sin 4\theta - 20\sin 2\theta \}$$

$$\cos^5 \theta \cdot \sin^2 \theta = -\frac{1}{2^n} \left\{ \sin 12\theta - 2 \sin 10\theta - 4 \sin 8\theta \right. \\ \left. + 10 \sin 6\theta + 5 \sin 4\theta - 20 \sin 2\theta \right\}$$

Answer

Q 2 B

$\cos^4 \theta$

Solution

$$(2 \cos \theta)^4 = \left(x + \frac{1}{x} \right)^4$$

$$= \frac{x^4 + 4x^3 + \frac{4 \cdot 3}{2!} x^2 \cdot \frac{1}{x^2} + \frac{4 \cdot 3 \cdot 2}{3!} x \cdot \frac{1}{x^3}}{+ \frac{4!}{4!} \cdot \frac{1}{x^4}}$$

$$= x^4 + 4x^2 + 6 + \frac{4}{x^2} + \frac{1}{x^4}$$

$$= x^4 + \frac{1}{x^4} + 4 \left(x^2 + \frac{1}{x^2} \right) + 6$$

$$2^4 \cdot \cos^4 \theta = 2 \cos 4\theta + 4 (2 \cos 2\theta) + 6$$

$$\cos^4 \theta = \frac{2}{2^4} (\cos 4\theta + 4 \cos 2\theta + 3)$$

$$\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$$

Answer

Q 3 BSin⁶θ

Solution

$$(2 \sin \theta)^6 = \left(x - \frac{1}{x} \right)^6$$

$$= x^6 + 6(x^5)\left(\frac{-1}{x}\right) + \frac{6 \cdot 5}{2!}\left(\frac{-1}{x}\right)^2 + \frac{6 \cdot 5 \cdot 4}{3 \cdot 2} x^3$$

$$\left(\frac{-1}{x}\right)^3 + \frac{6 \cdot 5 \cdot 4 \cdot 3}{4 \cdot 3 \cdot 2} x^2 \left(\frac{-1}{x}\right)^4 + \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{5!} x$$

$$\left(\frac{-1}{x}\right)^5 + \frac{6!}{6!} \left(\frac{-1}{x}\right)^6$$

$$= x^6 - 6x^4 + 15x^2 - 20 + \frac{15}{x^2} - \frac{6}{x^4} + \frac{1}{x^6}$$

$$= x^6 + \frac{1}{x^6} - 6\left(x^4 + \frac{1}{x^4}\right) + 15\left(x^2 + \frac{1}{x^2}\right) - 20$$

$$2^6 \cdot \sin^6 \theta = \frac{-1}{25} (\cos 6\theta - 6 \cos 4\theta + 15 \cos 2\theta - 10)$$

Answer

QUESTION

Prove the following

$$\underline{\underline{\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}}$$

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

$$\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

Solution

we know that

$$\therefore \cos n\theta + i \sin n\theta = (\cos \theta + i \sin \theta)^n$$

$$\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$$

expanding R.H.S by $(a+b)^n = a^n + n a^{n-1} b + \dots$

$$\Rightarrow \cos^4 \theta + i \sin^4 \theta = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta + \frac{4 \cdot 3}{2} \cos^2 \theta i^2 \sin^2 \theta + \frac{4 \cdot 3 \cdot 2}{3 \cdot 2} \cos \theta i^3 \sin^3 \theta + \frac{4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2} i^4 \sin^4 \theta$$

$$\Rightarrow \cos^4 \theta + i \sin^4 \theta = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

equating real and imaginary parts

Real Part

$$\cos^4 \theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$\cos^4 \theta = \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)$$

$$\cos^4 \theta = \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$\cos^4 \theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \rightarrow \text{eq A}$$

Imaginary part

$$\sin^4 \theta = 4 \cos^3 \theta i \sin \theta - 4 \cos \theta i \sin^3 \theta \rightarrow \text{eq B}$$

From eq A and eq B

Proved